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Post-hoc Tests Explained: Tukey, Bonferroni, Holm-Bonferroni and Scheffé's test

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Note: This article builds on my previous article on one-way ANOVA. Please read it if you want a refresher on how the one-way ANOVA works

Country, Other	Total Cases	New Cases	Total Deaths	New Deaths	Total Recovered	Active Cases	Serious, Critical
China	80,881	+21					
Italy	31,506	+3,526	3,226	+13	68,715	8,940	3,226
Iran	16,169	+1,178	2,503	+345	2,941	26,062	2,060
Spain	11,409	+1,467	988	+135	5,389	9,792	
Germany	8,639	+1,367	510	+168	1,028	9,871	563
S. Korea	8,320	+1,367	23	+6	67	8,549	2
France	8,320	+84	81	+6	1,401	6,838	59
USA	6,633		148		12	6,473	400
Switzerland	5,704	+1,041	97	+11	74	5,533	12
UK	2,742	+389	27	+8	15	2,700	
Netherlands	1,950	+407	71	+16	52	1,827	20
Norway	1,705	+292	43	+19	2	1,660	46
Austria	1,452	+104	3		1	1,448	27
Belgium	1,332	+314	4	+1	8	1,320	12
Sweden	1,243	+185	10		14	1,219	33
Denmark	1,191	+70	8	+1	1	1,182	12
Japan	977	+63	4		1	972	18
Diamond Princess	878	+45	29	+1	144	705	41
Malaysia	696		7		456	233	15
Australia	673	+107	2	+2	49	622	10
Canada	673	+107	2	+2	27	420	1
Portugal	452	+51	5	+1	11	436	1
Qatar	452	+11	5	+1	3	444	18
Czechia	448	+117	1		4	435	2
Greece	439				3	393	11
Israel	396	+52	5	+1	14	368	5
Finland	387	+35			11	313	2
Brazil	387	+26			10	312	18
	324	+44	1	+1	2	298	
	322	+67	1			274	
	301	+22					

Photo by [Christine Sandu](#) on [Unsplash](#)

You just performed an ANOVA test, and it came back significant. *That's exciting, but now what?* The ANOVA tells you that at least one group differs, but it doesn't reveal which ones. You can only conclude that one group is significantly different from the other, but it would be interesting to know which group isn't? That is where the *post-hoc test* comes in handy and will help us to compare group pairs and uncover exactly where the difference lies! In this article, I will explain what post-hoc tests are and present the most popular ones, using the same example as my [previous article](#) on one-way ANOVAS

Post-hoc tests

The post-hoc tests allow us to compare groups in an ANOVA, while reducing what we call the **family-wise error rate (FWER)**. The FWER is the probability of making one or more Type I errors (false positives) when performing multiple hypothesis tests. That is the case because when we conduct several hypothesis tests at once using a significance level of .05, the probability that we get a false positive increases beyond just 0.05.

$$FWER = 1 - (1 - \alpha)^m$$

Family-wise error rate formula

Where :

- m = the number of tests
- α = the significance level of each test (for a review of the alpha, you can read [this article](#))

To illustrate the concept of FWER, imagine we want to compare four groups together (Group 1, Group 2, Group 4 and Group 4) in the following way :

- Group 1 — Group 2
- Group 1 — Group 3
- Group 1 — Group 4
- Group 2 — Group 3

- *Group 2 — Group 4*
- *Group 3 — Group 4*

We have six pairwise comparisons to calculate. Over time, the FWER increases quickly, as we can see from that table :

<u>Number of groups</u>	<u>Number of comparison</u> $Comparisons = \frac{(Groups * (Groups - 1))}{2}$	<u>FWER</u> $FWER = 1 - (1 - \alpha)^m$
3	3	0.1426
4	6	0.2649
5	10	0.4013
6	15	0.5367
7	21	0.6594
8	28	0.7622
9	36	0.8422
10	45	0.9006

With 10 groups to compare, we now have a 90% risk of making a Type 1 error, which is pretty huge. However, post-hoc tests allow us to reduce the risk of FWER considerably.

In this article, we'll explore the most commonly used post-hoc tests, when to use them, and how to interpret their results. Whether you're analysing marketing campaigns, clinical trials, or survey responses, post-hoc tests are essential to make sense of your data after performing an ANOVA.

Popular Post-Hoc Tests

In order to walk you through different post-hoc tests, I will use the same example as my previous article. We have the following groups and means :

Group	Mean Score
A (Textbook)	72
B (Videos)	78
C (Flashcards)	85

Where :

- **Group A** : Reads the textbook only (mean = 72; SD = 5; n = 30)

- **Group B** : Watches video lectures (mean = 78; SD = 5; n = 30)
- **Group C** : Practices with flashcards (mean = 85; SD = 5; n = 30)

I will run a python script to simulate a dataset and perform a one-way ANOVA on it :

```
import numpy as np
import pandas as pd
from scipy import stats
import statsmodels.api as sm
from statsmodels.formula.api import ols
from statsmodels.stats.multicomp import pairwise_tukeyhsd
import pingouin as pg

# Simulate the data
np.random.seed(42)
n = 30
group_A = np.random.normal(loc=72, scale=5, size=n)
group_B = np.random.normal(loc=85, scale=5, size=n)
group_C = np.random.normal(loc=75, scale=5, size=n)

# Create DataFrame
df = pd.DataFrame({
    'score': np.concatenate([group_A, group_B, group_C]),
    'group': ['Textbook']*n + ['Videos']*n + ['Flashcards']*n
})

# One-way ANOVA
model = ols('score ~ group', data=df).fit()
anova_table = sm.stats.anova_lm(model, typ=2)
print("🔍 ANOVA Result:\n", anova_table)
```

When running our ANOVA, we get the following result: $F(2, 87) = 66.54$, $p = 0.0$.
Again, to have an overview of how I got there, [please review this article](#).

Using these results, I will show 4 post-hoc test that can be used, such as

- Tukey's HSD
- Bonferonni
- Holm-Bonferonni

- Scheffé's

1. Tukey's HSD

The Tukey Honest Significant Difference (HSD) test helps us determine if there are significant differences between the means of our groups by comparing all possible pairs of means while controlling for the FWER. The formula for this test is the following :

$$q_s = \frac{|\bar{X}_a - \bar{X}_b|}{SE}$$

Tukey's HSD formula

where :

- q_s : The *Studentized range statistic* (used to compare group means)
- X_a and X_b : Sample means of the two groups being compared
- SE : Standard error of the difference between means

The Studentized range distribution (often denoted as “ q_s ” or “ q ”) describes the probability distribution of the range of sample means, scaled by the standard error. It's used when comparing multiple group means to determine if the largest and smallest differ significantly. This distribution is useful for determining the probability of observing a range (difference between the largest and smallest sample means) that has been standardised (or “studentized”) by dividing by an estimate of the standard error (our SE in this example) :

$$SE = \frac{s}{\sqrt{n}}$$

Standard Error formula

Where :

- s = Standard deviation
- n = Sample size

How to interpret Tukey's HSD

```
# Tukey's HSD
tukey = pairwise_tukeyhsd(endog=df['score'], groups=df['group'], alpha=0.05)
print("\n📊 Tukey's HSD:\n", tukey)
```

```
ANOVA Result:
F-statistic: 66.5409, p-value: 0.0000

Tukey's HSD Result:
Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====
group1    group2    meandiff    p-adj    lower    upper    reject
-----
Flashcards Textbook -14.0052    0.0    -16.9044  -11.106    True
Flashcards  Videos  -7.6702    0.0    -10.5694  -4.771     True
Textbook   Videos   6.3349    0.0     3.4357    9.2341     True
-----
```

Looking at this data, we can conclude that all the comparisons are significant (because all p-values are below our threshold of 0.05). We can also see by looking at the **meandiff** column that *Flashcard* are statistically significant from *Textbook* by a margin of 14.01 (difference between the two means) and *Videos* by a margin of 7.67, and *Textbook* is significantly different from *Videos* by a margin of 6.33.

You could tell me that these results were obvious just by looking at the means in the table above (*yes... I know*). But this simple example is just to explain how Tukey's HSD works. Imagine a scenario where the means and standard deviation would be different, or that there would be way more groups, that's where post-hoc tests would be more useful.

2. Bonferroni

The second test I will present is the **Bonferroni test**. The way it works is by adjusting the significance level (alpha) for multiple comparisons to control for the probability of making a Type I error :

$$\alpha_{\text{adjusted}} = \frac{\alpha}{m}$$

Where:

- α : the original significance level (e.g., 0.05)
- m : number of comparisons
- α_{adjusted} : the new threshold for each individual test

Basically, Bonferroni will increase our α (and our p-value) the more comparisons we have. The problem with that test is that it becomes *veeeery* conservative the more groups we are comparing, and we are increasing our risks of doing a Type II error (false negative) where we think there isn't an effect when there actually is one!

How to interpret Bonferroni

So if we take our three groups (Textbooks, Videos, Flashcards), by applying the Bonferonni formula, we would have the following α :

$$\alpha_{\text{Bonferroni}} = \frac{\alpha}{m} = \frac{0.05}{3} \approx 0.0167$$

This means that for each individual test, you'd only consider a result statistically significant if its p-value is below **0.0167**, not the usual 0.05.

ANOVA Result:

F-statistic: 66.5409, p-value: 0.0000

Tukey's HSD Result:

Multiple Comparison of Means - Tukey HSD, FWER=0.05

group1	group2	meandiff	p-adj	lower	upper	reject
Flashcards	Textbook	-14.0052	0.0	-16.9044	-11.106	True
Flashcards	Videos	-7.6702	0.0	-10.5694	-4.771	True
Textbook	Videos	6.3349	0.0	3.4357	9.2341	True

By taking our results above, all our p-values from the *p-adj* column are statistically significant (0.0) because they are still smaller than our new threshold ($\alpha = 0.0167$) set by the Bonferroni test.

Bonferroni is by far the simplest post-hoc test. However, it can be problematic since it can greatly reduce the statistical power of our test by being overtly conservative (and increasing the risk of doing a type II error), the Bonferroni method, and also treats all the comparisons equally. There exists a less conservative alternative called the Holm-Bonferroni method introduced by the mathematician Sture Holm in 1979 to adjust those issues.

3. Holm-Bonferonni

Since Bonferroni is pretty conservative, there is an alternative to it called Holm-Bonferroni (*clever, right?*). The Holm-Bonferroni post hoc test works by adjusting the significance of the p-value according to its size. Instead of adjusting the α to a new threshold, it will do it differently for all comparisons, according to this formula :

$$p_{(i)} \leq \frac{\alpha}{m - i + 1} \quad \text{for } i = 1, 2, \dots, m$$

where :

- p_i : the i -th smallest p -value
- m : total number of hypotheses
- α : overall significance level (e.g., 0.05)

Basically, the Holm procedure works by first ordering all p -values from smallest to largest ($p_1 \leq p_2 \leq \dots \leq p_m$). The smallest p -value (p_1), compared against α/m . If $p_1 < \alpha/m$, we can reject the corresponding null hypothesis and continue; otherwise, stop and accept all remaining hypotheses. For the second p -value (p_2), compare against $\alpha/(m-1)$ and then continue this process, comparing each p -value p_i against $\alpha/(m-i+1)$!

How to interpret Holm-Bonferroni

So let's say that we have 4 tests (A, B, C, D) with the following p -values:

- $A = 0.003$
- $B = 0.004$
- $C = 0.026$
- $D = 0.037$

We can calculate the new p -value to compare with the following formula (notice that for each p -value, we increase the denominator by 1) :

$$[p_{(1)} = 0.003 \leq \frac{0.05}{4} = 0.0125 \Rightarrow \text{Reject } H_{(1)}]$$

$$[p_{(2)} = 0.004 \leq \frac{0.05}{3} \approx 0.0167 \Rightarrow \text{Reject } H_{(2)}]$$

$$[p_{(3)} = 0.026 \not\leq \frac{0.05}{2} = 0.025 \Rightarrow \text{Do not reject } H_{(3)}]$$

$$[p_{(4)} = 0.037 \leq \frac{0.05}{1} = 0.05 \Rightarrow \text{Do not reject } H_{(4)}]$$

If we compare the results we get from Holm-Bonferroni with Bonferroni, we get

this table :

Test	P-value	Bonferroni threshold ($\alpha=0.05$)	Holm threshold ($\alpha=0.05$)	Bonferroni result	Holm result
A	0.003	0.01	0.0125	Significant	Significant
B	0.004	0.01	0.0167	Significant	Significant
C	0.026	0.01	0.025	Not significant	Significant
D	0.037	0.01	0.05	Not significant	Not significant

As you can see, with Holm-Bonferroni, for the test C, since our threshold is different (0.025 instead of 0.01), we are suddenly much closer to accept the result of the test, compared to Bonferroni.

Okay now, onto the last test for today (*of course, there is more post hoc test, but the article would get a bit too long if I listed them all!*)

4. Scheffé's test

The final test I will present to you today is the **Scheffé test**, which is used when you want to make multiple comparisons between group means, especially **custom contrasts**, not just pairwise ones. It's known for being **very conservative**, which means it strongly controls the risk of Type I errors (false positives), but may reduce statistical power. It is often used when our sample size between our groups are unequal.

Scheffé's test uses the F-distribution to determine whether a contrast is statistically significant. The critical value is adjusted to account for the fact that many comparisons are being made.

To perform it, we calculate a contrast between our sample mean :

$$F = \frac{C^2}{\left(\sum_{i=1}^k \frac{c_i^2}{n_i} \right) \cdot MSE}$$

Where:

- C : Estimated contrast using sample means
- n_i : Sample size of group i
- MSE : Mean square error from the ANOVA
- k : Nnumber of groups

We then compare the F we obtained above using the MSE and the contrast with a critical F value :

$$F_{\text{critical}} = (k - 1) \cdot F_{\alpha, k-1, N-k}$$

Where:

- F_{critical} : Critical value from the F -distribution at significance level α
- N is the total number of observations

How to interpret Scheffé's test

Suppose we want to compare the average of Groups C (Flashcards) and B (videos) vs. the average of Groups A (Textbooks) and C (Flashcards). We're comparing Group C (mean = 85) vs. the average of Group A (mean = 72) and Group B (mean = 78)

We can assign different contrasts to our values, such that (*note: this will change depending on the comparison you want to do, but I'm not so sure how... for more about contrast in the context of post-hoc test, you can read [this article here](#)*) :

- $c_A = -0.5$
- $c_B = -0.5$
- $c_C = +1$

From there, we calculate our contrast C like this :

$$C = -0.5 \cdot 72 - 0.5 \cdot 78 + 1 \cdot 85 = -36 - 39 + 85 = 10$$

and the denominator (c / n) :

$$\sum \frac{c_i^2}{n_i} = \frac{(-0.5)^2}{30} + \frac{(-0.5)^2}{30} + \frac{(1)^2}{30} = \frac{0.25}{30} + \frac{0.25}{30} + \frac{1}{30} = \frac{1.5}{30} = 0.05$$

to then calculate our F :

$$F = \frac{C^2}{\left(\sum \frac{c_i^2}{n_i}\right) \cdot MSE} = \frac{10^2}{0.05 \cdot 25} = \frac{100}{1.25} = 80$$

Which we will compare with a critical value (F-scheff) :

$$F_{\text{Scheff}} = (k-1) \cdot F_{\alpha, k-1, N-k} = 2 \cdot F_{0.05, 2, 87} \approx 2 \cdot 3.10 = 6.20$$

Since our computed F ($F = 80$) is way bigger than the critical F ($F_{\text{Scheff}} = 6.20$), the contrast is **statistically significant**. That means the flashcard group **significantly outperforms** the combined textbook and video lecture groups. This confirms that the flashcard group significantly outperforms the combined textbook and video lecture groups.

Best use case

I've made a little table to summarise the best post-hoc test to use. It isn't exhaustive since there are other tests out there, but it can still give an overview of the concepts that we talked about throughout the article.

Test	Best used when	Strength	Limitations
Tukey's HSD	You want to compare all possible pairs of means	Good balance of Type I error control and power	Less suitable for unequal variances or sample sizes
Bonferroni	You have a small number of planned comparisons (i.e, less than 3)	Simple and conservative; controls family-wise error rate	Can be overly conservative; reduced power
Holm-Bonferroni	You want a less conservative alternative to Bonferroni	Sequential adjustment; more powerful than Bonferroni	Slightly more complex to compute manually
Scheffé	You want to test any linear combination of group means (custom contrasts)	Extremely flexible; strong control of Type I error	Very conservative; lower power for pairwise comparisons

Conclusion

Whether you're comparing means with Tukey's HSD, adjusting for multiple comparisons with Bonferroni, or diving into different types of contrasts with Scheffé, the choice of test depends on your research goals, sample structure, and tolerance for Type I error.

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